1

Measurement Ready

Use the tool, scale, unit, and calculation as one evidence chain.

Name: ___________________ Date: _________ Class Period: ______

LEARNING TARGETS

Match common lab tools to the quantity and unit they measure. • Choose an appropriate tool and justify the choice with visual evidence. • Calculate density and percent error with correct units and reasonable precision.

Three tools, three different jobs

Electronic balance
Measures mass
Typical unit: g

Graduated cylinder
Measures liquid volume
Typical unit: mL

Beaker
Holds and mixes
Volume is approximate

Measurement rules that prevent avoidable errors

- Record the value and the unit together.
- For an analog scale, record all certain digits plus one estimated digit.
- Read a concave liquid meniscus at eye level from its lowest point.
- Keep units through every calculation and round only the final result.

FORMULA TOOLBOX

density = mass / volume
percent error =
|experimental - accepted| / accepted × 100

UNITS

mass: g
liquid volume: mL
density: g/mL

2

Choose the Tool

Name what you see, then use visible features to justify the choice.

Name: __________________ Date: _________ Class Period: ______

A. Identify each instrument

1

Instrument: _________________
Quantity/job: _______________
Unit: ______________________

2

Instrument: _________________
Quantity/job: _______________
Unit: ______________________

3

Instrument: _________________
Quantity/job: _______________
Unit: ______________________

B. Select and justify

4. You need 18.6 mL of water. Choose a beaker or a graduated cylinder. Explain which visible design features support your choice.

____

____

5. You need about 125 mL of solution and room to stir. Choose a beaker or a graduated cylinder. Explain.

____

____

C. Read visual evidence

6. Identify two visible features that show this vessel was designed to measure liquid volume.

____

____

Photo: Darrien / Haltopub, Public Domain, Wikimedia Commons

3

Calculate with Units

Show the formula, substitute values with units, and round only the final answer.

Name: __________________ Date: _________ Class Period: ______

GUIDED EXAMPLE

A sample has a mass of 36.4 g and a volume of 14.0 mL.

d = m / V = 36.4 g / 14.0 mL = 2.60 g/mL

7. A liquid sample has a mass of 27.0 g and a density of 2.70 g/mL. Calculate its volume.

Work space

8. A liquid has a density of 0.789 g/mL and a volume of 25.0 mL. Calculate its mass.

Work space

9. A student measures 2.58 g/mL. The accepted value is 2.70 g/mL. Calculate percent error.

Work space

10. Accuracy and precision

Accepted density: 2.70 g/mL

Set A	2.70, 2.71, 2.69 g/mL
Set B	2.42, 2.41, 2.42 g/mL

Which set is accurate? Which is precise? Explain using the pattern in the data.

____

____

4

Reason from Evidence

Use the data to build a claim, diagnose the error, and plan a repair.

Name: ________________ Date: ________ Class Period: _____

LAB CASE: UNKNOWN LIQUID

A student uses an electronic balance and a graduated cylinder. The empty container has a mass of 42.31 g. The container plus liquid has a mass of 67.19 g. The liquid volume is 20.0 mL. The accepted density is 1.25 g/mL.

11. Mass of liquid: __________________

12. Experimental density: __________________

13. Percent error: __________________

14. Claim - Evidence - Reasoning

Claim: Is the experimental result close to the accepted value?

____

Evidence: Cite at least two numbers from the case.

____

Reasoning: Explain what percent error tells you and what it does not tell you about precision.

____

____

15. Error diagnosis

Another student reads the liquid above eye level, rounds every intermediate value, and omits units. Name two problems and one exact repair action.

____

____

Item/skill to repair: _______________

Next action: ______________________

Incorrect idea or step: _____________

When I will retry: _________________

1

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Typical unit: g

Graduated cylinder

Measures liquid volume

Typical unit: mL

Beaker

Holds and mixes

Volume is approximate

Measurement rules that prevent avoidable errors

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- For an analog scale, record all certain digits plus one estimated digit.
- Read a concave liquid meniscus at eye level from its lowest point.
- Keep units through every calculation and round only the final result.

FORMULA TOOLBOX

density = mass / volume

percent error =

|experimental - accepted| / accepted × 100

UNITS

mass: g

liquid volume: mL

density: g/mL

2

Choose the Tool

Name what you see, then use visible features to justify the choice.

TEACHER KEY

A. Identify each instrument

1

Electronic balance
mass
g

2

Graduated cylinder
liquid volume
mL

3

Beaker
holding/mixing; rough volume only

B. Select and justify

4. You need 18.6 mL of water. Choose a beaker or a graduated cylinder. Explain which visible design features support your choice.

Graduated cylinder. Its narrow shape and graduated scale support a more precise volume reading than a beaker.

5. You need about 125 mL of solution and room to stir. Choose a beaker or a graduated cylinder. Explain.

Beaker. The task asks for an approximate amount and mixing space, not a precise delivered volume.

C. Read visual evidence

6. Identify two visible features that show this vessel was designed to measure liquid volume.

Any two visible features: tall/narrow vessel, graduated marks, numbered scale, stable base, or pouring lip.

Photo: Darrien / Haltopub, Public Domain, Wikimedia Commons

3

Calculate with Units

Show the formula, substitute values with units, and round only the final answer.

TEACHER KEY

GUIDED EXAMPLE

A sample has a mass of 36.4 g and a volume of 14.0 mL.
d = m / V = 36.4 g / 14.0 mL = 2.60 g/mL

7. A liquid sample has a mass of 27.0 g and a density of 2.70 g/mL. Calculate its volume.

10.0 mL

8. A liquid has a density of 0.789 g/mL and a volume of 25.0 mL. Calculate its mass.

19.7 g

9. A student measures 2.58 g/mL. The accepted value is 2.70 g/mL. Calculate percent error.

4.44%

10. Accuracy and precision

Accepted density: 2.70 g/mL

Set A	2.70, 2.71, 2.69 g/mL
Set B	2.42, 2.41, 2.42 g/mL

Which set is accurate? Which is precise? Explain using the pattern in the data.

Set A is accurate and precise. Set B is precise but not accurate.

4 Reason from Evidence

Use the data to build a claim, diagnose the error, and plan a repair.

TEACHER KEY

LAB CASE: UNKNOWN LIQUID

A student uses an electronic balance and a graduated cylinder. The empty container has a mass of 42.31 g. The container plus liquid has a mass of 67.19 g. The liquid volume is 20.0 mL. The accepted density is 1.25 g/mL.

11. Mass of liquid:
24.88 g

12. Experimental density:
1.24 g/mL

13. Percent error:
0.480% (0.48% acceptable)

14. Claim - Evidence - Reasoning

Claim: Is the experimental result close to the accepted value?

Evidence: Cite at least two numbers from the case.

Reasoning: Explain what percent error tells you and what it does not tell you about precision.

The result is close to the accepted value, but the student should repeat trials before judging precision. Evidence: 1.244 g/mL before final rounding vs. 1.25 g/mL and 0.48% error.

15. Error diagnosis

Another student reads the liquid above eye level, rounds every intermediate value, and omits units. Name two problems and one exact repair action.

Read the bottom of the meniscus at eye level, record one estimated digit, keep units through every step, and repeat measurements.

Item/skill to repair: ________________

Next action: ________________________

Incorrect idea or step: ______________

When I will retry: __________________